

NAME: _____

The Copper Kettle Loan Decision

a production-and-financing decision • Linear Algebra

Nadia Reyes runs Copper Kettle Foods, a two-product bakehouse in Dayton: her classic stovetop granola and a cherry-almond snack bar, sold through a Saturday farmers-market stall and a handful of small grocers. She wants a second oven, and Riverline Community Bank will underwrite the equipment loan only if Copper Kettle can document that its best achievable weekly operating profit is at least \$1,500. Nadia's business advisor set up the planning grid below to find that best profit, ran out of session time, and left a one-page procedure sheet for finishing the job. Nadia's cousin Marco has already finished the update his own way, and he wants the loan application submitted today. Your group gets the final word.

The week at Copper Kettle

- Granola: 2 oven-hours and 2 packing-hours per batch; \$90 profit per batch.
- Cherry-almond bars: 1 oven-hour and 5 packing-hours per batch; \$40 profit per batch.
- Available each week: 24 oven-hours and 40 packing-hours.
- Bank requirement: documented best achievable weekly profit of at least \$1,500.
- In the grid: g and c are batches of granola and bars; u_1 and u_2 are the advisor's bookkeeping quantities, recording a plan's *unused* oven-hours and *unused* packing-hours; P is weekly profit in dollars.

The advisor's planning grid (as she left it)

	g	c	u_1	u_2	P	Week
Oven row	2	1	1	0	0	24
Packing row	2	5	0	1	0	40
Profit row	-90	-40	0	0	1	0

The advisor's procedure sheet (treat these rules as given)

1. Find the most negative number in the bottom row. Its column is this round's *work column*.
2. For each row above the bottom one whose work-column entry is positive, divide the last-column number by that entry. The row with the *smallest* quotient is the *work row*.
3. Scale the work row so its work-column entry becomes 1, then add suitable multiples of the work row to every other row, the bottom one included, until every other entry in the work column is 0.
4. Reading a plan from a grid: a column holding a single 1 with 0s elsewhere is *settled*, and its quantity equals the last-column entry of the row containing the 1. Every quantity whose column is not of that form equals 0.
5. Stopping check: when no bottom-row entry under the four quantity columns (g , c , u_1 , u_2) is negative, no further round can raise profit, and the bottom row's last-column entry is the

largest weekly profit any plan within the week's hours can achieve.

1. Run One Update Round

Following the procedure sheet exactly, carry out one full update round on the advisor's grid. State which column you worked on and which rule chose it, show the two quotients that chose your row, and write out the complete updated grid, bottom row included. *Hint: Rules 1 and 2 make both choices for you; rule 3 is row arithmetic you have done in class.*

2. Read Off the Week's Plan

Using rule 4, read the plan your updated grid describes: batches of granola, batches of bars, unused oven-hours, unused packing-hours, and the week's profit. Then say in one sentence what this plan tells Nadia to do with the bar line. *Hint: Check each column against rule 4 before you write any number down.*

3. Is This as Good as It Gets?

Apply rule 5 to your updated grid. Could another update round raise the week's profit? Point to the exact entries that justify your answer, and state the largest weekly profit any plan within the week's hours can achieve. *Hint: The answer lives entirely in the bottom row; check the sign of every entry under the four quantity columns.*

4. Settle Marco's Number

At rule 2, Marco chose the packing row instead, "because 40 hours leaves more room than 24," and his finished grid reads a weekly profit of \$1,800, comfortably over the bank's bar. As a group, decide who is right: find the specific number in Marco's grid that cannot be true of any real production week, and then make the call against the bank's \$1,500 requirement. *Hint: Before trusting any profit figure, check the plan behind it against the week's oven-hours.*

Group recommendation. Write Nadia 2 to 3 sentences: submit the loan application this week or hold it, citing the one profit figure that survives your checks and what would have to change before you would revisit.

Instructor Material

Analyst's Checklist (optional, for stuck groups)

1. Circle the most negative bottom-row entry; its column is the work column.
2. Form both quotients (last column divided by work-column entry); the smaller one picks the work row.
3. Scale the work row first; then clear the work column's other two entries one row at a time, writing each row operation down before computing it.
4. Before reading any plan, mark every column settled or not settled per rule 4.
5. For the stopping check, scan only the signs in the bottom row under g , c , u_1 , u_2 .
6. Compare the one surviving profit figure against \$1,500 before drafting the recommendation.

Answer Key

1. Run One Update Round *Skill: Executing one simplex pivot: entering column by the most-negative objective-row entry, leaving row by the minimum-ratio test, pivot via elementary row operations.*
Work column: g , chosen by rule 1 since -90 is the most negative bottom-row entry (compare -40). Quotients (rule 2): oven row $24/2 = 12$; packing row $40/2 = 20$; smallest is 12, so the oven row is the work row. Row operations: $R_1 \rightarrow \frac{1}{2}R_1$, then $R_2 \rightarrow R_2 - 2R_1$, then $R_3 \rightarrow R_3 + 90R_1$.

	g	c	u_1	u_2	P	Week
Oven row	1	$\frac{1}{2}$	$\frac{1}{2}$	0	0	12
Packing row	0	4	-1	1	0	16
Profit row	0	5	45	0	1	1080

2. Read Off the Week's Plan *Skill: Reading the basic feasible solution off a tableau: unit columns are basic, all other variables are zero.*

Settled columns: g (unit in the oven row), u_2 (unit in the packing row), and P . Columns c and u_1 are not settled, so those quantities are 0.

$$g = 12, \quad c = 0, \quad u_1 = 0, \quad u_2 = 16, \quad P = \$1,080$$

Interpretation: bake 12 batches of granola and pause the bar line entirely; the oven is used to the last hour ($u_1 = 0$) while 16 packing-hours sit idle; the week clears \$1,080. Consistency check in scenario terms: $2(12) = 24$ oven-hours (all used) and $2(12) = 24$ packing-hours ($40 - 24 = 16$ unused).

3. Is This as Good as It Gets? *Skill: Applying the simplex optimality criterion to the objective row.*

Bottom-row entries under g , c , u_1 , u_2 are 0, 5, 45, 0: none is negative, so by rule 5 no further round can raise profit.

$$P_{\max} = \$1,080$$

No plan within 24 oven-hours and 40 packing-hours earns more.

4. Settle Marco's Number (trap, resolved) *Skill: Minimum-ratio rule as a feasibility guard; comparing a valid and an invalid conclusion against the decision threshold.*

Marco pivoted on the packing row (quotient 20) instead of the oven row (quotient 12), violating rule 2. His row operations, $R_2 \rightarrow \frac{1}{2}R_2$, $R_1 \rightarrow R_1 - 2R_2$, $R_3 \rightarrow R_3 + 90R_2$, produce

	g	c	u_1	u_2	P	Week
Oven row	0	-4	1	-1	0	-16
Packing row	1	$\frac{5}{2}$	0	$\frac{1}{2}$	0	20
Profit row	0	185	0	45	1	1800

The impossible number is $u_1 = -16$: his plan claims *negative sixteen* unused oven-hours. Equivalently, his plan bakes $g = 20$ batches, which needs $2(20) = 40$ oven-hours when only 24 exist. Rule 2 exists precisely to prevent this: the smaller quotient (12) is the most the oven can actually support. Both numbers, side by side: Marco's \$1,800 (from an impossible week) versus the honest \$1,080. The gap decides the case: \$1,800 > \$1,500 says apply, but \$1,080 < \$1,500 says wait.

Hold the application: \$1,080 < \$1,500

Verdict. Do not submit this week. The only defensible figure is \$1,080, which misses the bank's \$1,500 requirement by \$420. A closing note groups often find on their own: the updated grid prices the bottleneck, since profit rises \$45 for each added oven-hour, so roughly nine and a half more weekly oven-hours would clear the bar. That is exactly what the second oven would provide, which is why Nadia should strengthen the business case before reapplying rather than submit an application the numbers cannot support.

Alignment

- **PRIMARY:** A First Course in Linear Algebra (Ken Kuttler), Section 1.3, "Gaussian Elimination." Grounds the row-operation mechanics of the update round (pivot positions, scaling, elimination).
- **SUPPORTING:** Understanding Linear Algebra (David Austin), Section 1.4, "Pivots and their influence on solution spaces." Grounds the rule-4 read-off: columns with pivot positions correspond to determined (basic) quantities, columns without to quantities set to zero.
- **SUPPORTING:** A First Course in Linear Algebra (Ken Kuttler), Section 1.2, "Systems of Equations, Algebraic Procedures." Grounds why the row moves are legitimate (elementary operations preserve the solution set).
- **Source problem learning objective:** Execute one iteration of the simplex method on a linear-programming tableau: identify the entering variable from the objective row, choose the pivot row by the minimum-ratio rule, perform the pivot via elementary row operations, read off the resulting basic feasible solution, and assess optimality from the bottom-row entries.
- **Section objectives exercised:** Kuttler 1.3, "Use Gaussian Elimination to solve systems of linear equations" (the round's row arithmetic); Austin 1.4, "columns of the coefficient matrix containing pivot positions correspond to basic variables and the columns without pivot positions correspond to free variables" (reading the plan); Kuttler 1.2, "Use elementary operations to find the solution to a linear system of equations" (validity of the update).
- **What makes this case distinct:** the trap is a ratio-rule violation whose phantom profit (\$1,800) and honest maximum (\$1,080) straddle the decision threshold (\$1,500), and the wrong grid is

refutable in scenario terms (negative unused oven-hours). Inert distractors: none among the printed numbers; flavor lives only in the prose (the farmers-market stall, Dayton).

- **Weak-mapping note (honest):** `lo_mapping` confidence is LOW (`primary_assessment` needs `change`, `supporting_assessment` incomplete, `multipart_assessment` `multipart_missing_parts`). No cited section teaches the simplex-specific rules (entering-column choice, minimum-ratio test, stopping criterion, plan read-off, slack interpretation); the worksheet supplies all five as reference data in the advisor's procedure sheet, with no technique named.
- **Attribution:** Adapted from “A First Course in Linear Algebra” by Ken Kuttler, used under CC BY 4.0. Adapted from “Understanding Linear Algebra” by David Austin, used under CC BY 4.0.